

Corporate social irresponsibility and stakeholder ecosystems: The case of Volkswagen Dieselgate scandal

Michael Zhang¹ | Glyn Atwal² | Maya Kaiser²

¹Nottingham Business School, Nottingham Trent University, Nottingham, United Kingdom

²Burgundy School of Business, Université Bourgogne Franche-Comté, Dijon, France

Correspondence

Michael Zhang, Nottingham Business School, Nottingham Trent University, 818 Newton Building, 50 Shakespeare Street, Nottingham, NG1 4FQ, United Kingdom.
Email: michael.zhang@ntu.ac.uk

Abstract

The Volkswagen emissions cheating scandal, known as “Dieselgate,” demonstrates that businesses can safeguard corporate social responsibility objectives and prevent or repair damage from corporate social irresponsibility by involving the stakeholder ecosystem in positive value co-creation. There is a lack of strategic management research offering a systematic approach to different stakeholders and their varied roles in responding to companies' corporate social irresponsibility. We adopt the theoretical lenses of stakeholder ecosystems and stakeholder value co-creation to re-examine Volkswagen's Dieselgate scandal to delineate the complex and evolving interactions among concerned stakeholders, which can shed light on the ongoing debate and research on brand management, corporate reputation, trust, sustainability, and sustainable mobility.

JEL CLASSIFICATION

L62; M11; M14; M48

1 | INTRODUCTION

Ethical transgressions by corporations are the damaging counterpart to corporate social responsibility (CSR). More than simple errors, these transgressions—which have been referred to as corporate social irresponsibility (CSI)—can have an enormous effect on a company's reputation, brand management, and its relationship with its stakeholders (Antonetti, 2020; Grappi, Romani, & Bagozzi, 2013; Lange & Washburn, 2012; Pearce & Manz, 2011; Riera & Iborra, 2017).

The emerging literature on CSI has mainly considered its impact on consumers (Antonetti, 2020; Grappi et al., 2013), which can be profound. However, management studies have rarely addressed the effects of CSI on other stakeholders and how these are complicated by the stakeholder ecosystem. Stakeholders include consumers, employees, regulators, policymakers, investors, and non-governmental organizations (NGOs).

Consider the Volkswagen emissions cheating case, known as “Dieselgate,” which attracted global media attention in September

2015. The company's decision to install secret software to cheat emissions tests directly affected consumer health and safety. The scandal, when it became known, sharply impacted Volkswagen's reputation and led to an immediate decline in the company's share price. In the longer term, however, the causes and effects were more complex, spilling over into the broader market, and eventually resulting in a shrinking market for diesel cars.

If one considers the only consumer response, the following question arises: why did a CSI case implicate Volkswagen, and Volkswagen alone, have such a significant effect on the market for diesel vehicles? As we demonstrate in this paper, one must examine the stakeholder ecosystem and consider how different stakeholder groups interact with one another. These stakeholder interactions across different markets, United States and Europe, are also crucial to explain what motivated Volkswagen's efforts to cheat emission's tests, and how the scandal broke when it did. By considering these interactions, we develop a systematic approach to the role of the stakeholder system through various stages of how CSI events develop.

2 | CSI

The concept of CSI is relatively recent in comparison to its more established antonym CSR (Carroll, 1979; McGuire, Sundgren, & Schneeweis, 1988; Murray & Vogel, 1997). Early studies of CSR focused primarily on companies' CSR actions and their social and financial performances (Carroll, 1979; Coffey & Fryxell, 1991). Marketing research has shown a positive correlation between a company's strategic CSR initiatives and consumer attitudes toward its reputation and products (Brown & Dacin, 1997). It is widely acknowledged that companies regard CSR initiatives not only as a response to external stakeholder pressure but also a pre-emptive strategy to gain a market advantage over their competitors (Brown & Dacin, 1997; Klein & Dawar, 2004; Sen & Bhattacharya, 2001).

As Aouad & Marsat (2018, p. 1027) point out, studies of CSR generally focus on positive social performance, but CSI can affect how a company performs even more significantly. Pearce & Manz (2011, p. 563) define CSI as "unethical executive behaviour that shows disregard for the welfare of others that at its extreme is manifested when executives seek personal gain at the expense of employees, shareholders and other organisation stakeholders, and even society at large." CSI can result in the collapse of an entire company, as in the Enron accounting scandal. Recent research has begun to pay greater attention to how consumers react to CSI events (Antonetti, 2020; Grappi et al., 2013), providing valuable insights into how CSI relates to anti-consumption (Kozinets & Handelman, 2004) or various types of unethical practices (Rindell et al., 2014). Indeed, CSI plays a central role in anti-consumption. As Lee et al. (2009) report, brand avoidance can be attributed to experience, identity, moral and deficit-value avoidance, while Bryson and Atwal (2018) have identified a relationship between brand hate and perceived irresponsible corporate behavior.

3 | VALUE CO-CREATION IN STAKEHOLDER ECOSYSTEMS

Vargo and Lusch (2008, p. 6) contend that value is "defined by and co-created with the consumer rather than embedded in output." The logic of value co-creation in this context considers the involvement of consumers in the creation of value primarily as stakeholders. Researchers have expanded the approach of value co-creation to include broader stakeholder ecosystems through their engagement in companies' innovation activities in the form of networks (Reyppens, Lievens, & Blazevic, 2016) and multi-stakeholder ecosystems (Kazadi, Lievens, & Mahr, 2016; Pera, Occhiocupo, & Clarke, 2016).

Gyrd-Jones and Kornum (2013) introduce the concept of stakeholder ecosystems in their discussion of brand value co-creation with consumers and other stakeholders. They posit that a stakeholder ecosystem is the system of interactions between the sociocultural players within a particular group of stakeholders. It encapsulates "both the network nature of these relationships and the complex set of subcultures that make up this ecosystem" (Gyrd-Jones & Kornum, 2013, p. 1484). They offer a critique of the company-centric approach to management by incorporating the concept of complementarity for

firms and their stakeholders to engage in interactive and reciprocal processes of value co-creation. Besides, stakeholder value co-creation, which has evolved as an emerging field of research (Kazadi et al., 2016; Reyppens et al., 2016) reinforces our understanding of the complex interactions between various stakeholders.

In the following, we look at how the complex interactions between stakeholders contributed to the unfolding of the Dieselgate scandal in three distinct phases.

3.1 | Dieselization in Europe (1998–2008)

The Volkswagen emissions cheating scandal occurred at a point when the demand for diesel cars was at its height in the European market. In the mid-1990s, diesel cars made up less than 10% of the automobile market in Europe. By 2001, that had grown to 36%. A decade later, in 2011, diesel cars made up more than half of all vehicles sold in Europe (55%) (Transport & Environment, 2018; Vidal, 2015). The rapid increase in diesel car sales during this period was driven by the European Commission's (EC) initiative in 1998 to reduce CO₂ emissions in response to the United Nations' call to mitigate climate change (European Commission, 1998). After EU members signed the Kyoto Protocol in December 1997, the European Union committed to reducing CO₂ emissions, and in July 1998 the EC reached an agreement with European carmakers to reduce CO₂ emissions from passenger cars. The automotive industry was a significant contributor to, as well as the key player in the reduction of, CO₂ emissions. The average CO₂ emission from cars in Europe was then 186 g/km. The European Automobile Manufacturers Association (ACEA) agreed with the EC to meet a target of 140 g/km from new passenger cars in order to achieve a 25% reduction of CO₂ emissions by 2008 (European Commission, 1998).

European regulations favored diesel cars over petrol cars, mainly because of their lower CO₂ emissions. Compared with petrol cars, diesel cars render about 20% higher fuel efficiency and 20% lower CO₂ emissions (Evans, 2020). Therefore, dieselization emerged as a cost-effective strategy for the ACEA to tackle the task of CO₂ emissions reduction. German carmakers and primarily Volkswagen invested in the development of advanced diesel engines in the 1980s. A major technological breakthrough was Volkswagen's innovative "turbocharged direct injection" engine (TDI), developed in 1989 (Miravete, Moral, & Thurk, 2015). This new engine, technically known as the EA189, was a "cleaner" diesel model with improved fuel efficiency and better performance in CO₂ emission's tests.

The new European regulations provided legal and financial incentives for carmakers and car dealerships to focus on diesel cars, and the TDI engine placed Volkswagen in an ideal position to emerge as a market leader. By 2008, capitalizing on its success in Europe, Volkswagen was ready to grow the market for diesel cars in the US, aiming "to set new environmental standards in vehicles, powertrains and lightweight construction" (Volkswagen, 2009).

3.2 | The United States market expands (2009–2015)

Volkswagen Group of America was established in 1955 as a wholly-owned subsidiary of Volkswagen AG (VW of America, 2020). In 2008, the Volkswagen group formulated the ambitious goal of becoming a global economic and environmental leader among automobile manufacturers through its “Strategy 2018” (Volkswagen, 2009). A critical objective of Strategy 2018 was to increase Volkswagen's share of the US diesel market. Stefan Jacoby, President and CEO of the Volkswagen Group of America, underlined Volkswagen's commitment to “becoming the most innovative high-volume brand in the US” (Noria Corporation, 2008).

Historically, in the 1970s and 1980s, US consumers had a negative perception of diesel cars (Lussenhop, 2015). Diesel engines lacked technological sophistication, and consumers had a general impression that diesel fuel was dirty. This perception started to change when the Volkswagen 2009 Jetta TDI won the Green Car Journal's 2009 Green Car of the Year at the LA Auto Show 2008 in Los Angeles, California. It was the first time that a diesel car model had won the award (Groom & Krolicki, 2008).

For diesel cars to be regarded as ‘green’ and to pass more stringent nitrogen oxide (NO_x) standards set by the US Environmental Protection Agency (EPA) reassured the US public. A key point in what would become the emissions cheating scandal is that regulations in the US and Europe targeted different types of car emissions. The European Emissions Standards Euro 1 – 4 between 1992 and 2004 (Automobile Association, 2020) was stricter on CO₂ emissions, requiring 30% lower CO₂ emissions than US Tier 1 Standards in the year 2000. However, the US was much stricter requiring NO_x emissions to be six times lower than they were permitted to be in the EU (Fisher, 2015; Miravete et al., 2015). The difference is that CO₂ has a more evident impact on climate change, while the EPA was “mainly concerned with emissions which are or would be harmful to people” (EPA, 2020). In particular, NO_x emissions can contribute to cardiovascular and respiratory diseases (Chossière et al., 2017).

Three sets of Emissions Standards, also called Federal Standards, Tier 1–3 standards, were introduced successively from 1994 to 2014 with the phase-in year of 1994, 2004, and 2014, respectively (Dieselnet, 2020). Volkswagen marketed “clean,” “green,” diesel models on the US market that ostensibly followed US regulations on NO_x emissions. Sustainability was at the core of its communications on corporate policy and strategy (Volkswagen, 2010). A group-wide CSR committee was created in 2009 to coordinate CSR-related activities across brands and regions. Alongside its vigorous marketing campaign, the company also registered with leading sustainability indices, including the Advanced Sustainability Performance Indices (ASPI), Dow Jones Sustainability World Index, and FTSE4Good Environmental Leaders Europe 40 (Volkswagen, 2010).

In 2014, the year before news broke of its cheating on emissions tests, Volkswagen delivered 586,700 passenger cars in the North America region, of which 63% were purchased by US consumers (VW US Media, 2015). The same year also saw Volkswagen's global

sales exceeding the ten million-unit mark, 3 years ahead of its ambitious Strategy 2018 goal. The Diesel Technology Forum cited a survey in 2015, reporting that 40% of US consumers would consider purchasing a diesel car, up from only 13% in 2005 (Lussenhop, 2015).

3.3 | The emissions cheating scandal emerges (2012–2015)

Volkswagen's success on the US market did, however, raise some concerns. In 2012, a group of US environmentalists queried why VW diesel car models driving on US roads supposedly generated less pollution than in Europe (McGee, 2017). The International Council on Clean Transportation (ICCT), an environmental NGO, started testing diesel cars on-road performance in 2013. The results showed that NO_x emissions from the tested cars were about 35 times the legal limit set by the EPA (McGee, 2017). However, when the cars were tested under laboratory conditions, the results met EPA standards.

In 2014, ICCT shared their results with both Volkswagen and the EPA for verification. The subsequent EPA investigation took more than a year to complete. Due to the sensitivity of the issue, the EPA was reluctant to make any public statement about the “incident” without firm evidence (McGee, 2017). In August 2015, however, the company's deliberate cheating on emission's tests was confirmed when a VW employee revealed that certain VW diesel cars had software installed that could detect laboratory tests and instruct the cars to enter a special lab mode where they produced lower emissions (McGee, 2017). The software of this kind is known as a “defeat device,” and it is not new, but was prohibited by the EU in 2007 (European Commission, 2017). Volkswagen officially acknowledged their illegal software to the EPA on September 3rd, 2015, and the EPA made this finding public on September 18th.

As a result of the investigation, the EPA ordered the recall of 482,000 diesel vehicles sold in the US, which were fitted with the “defeat device,” dating back to the 2009 model (Wright & Sharman, 2015). In Europe, an estimated 8.5 million Volkswagen diesel cars had the defeat device installed (McGee, 2016). To date, legal costs and fines related to the scandal have cost Volkswagen more than 30 billion euros (BBC News, 2020). The human cost of the scandal was also high: drawing on German data, Chossière et al. (2017) estimate that the excess emissions Volkswagen tried to cover up led to 1,200 premature deaths in Europe, with associated costs of 1.9 billion euros.

4 | DISCUSSION

A single factor did not drive the Volkswagen emissions cheating scandal. It was the result of complex interactions between different stakeholders who shaped Volkswagen's strategy on the automobile market. These can best be understood by looking at both stakeholder ecosystems (Gyrd-Jones & Kornum, 2013) and value co-creation (Kazadi et al., 2016; Pera et al., 2016), summarized in Table 1.

TABLE 1 Value co-creation in VW's stakeholder ecosystems

VW's stakeholder ecosystem	Value co-creation		
	Dieselization in Europe	Strategy 2018	Dieselgate scandal
Management	Technological innovation and market competition	US market expansion + sustainability campaigns	Ineffective compliance response to misconduct allegations
Employees	Innovation in TDI technology	Marketing the diesel cars	Internal whistle-blowers revealing the scandal
EU regulators	Proving a regulatory regime that incentivized dieselization with a focus on CO ₂ reduction to tackle global climate change		
ACEA	Committed to CO ₂ reduction with EC by lobbying dieselization of part of car fleet		
US regulators			Investigation and punishment through the discovery of VW's cheating device on NO _x emission with a focus on human health impact
Consumers	Positive consumer acceptance of diesel cars	Persuaded that diesel cars are "clean" and "green"	Seeking compensation and evidence of anti-brand behavior
Car dealership	Part of marketing activities in Europe	Part of marketing activities in the US	
Media	Positive coverage in Europe on the new "clean" diesel engines	Positive coverage in the US on VW's new "clean" engine and its commitments to CSR and sustainability reporting	Hesitant to cover the scandal before EPA's verdict
NGOs			External whistle-blowers exposing use of "cheat device"

The rise of diesel cars in Europe could not have occurred without the EC's commitment to reducing CO₂ emissions and its agreement with car manufacturers starting in 1998. During the dieselization period, the key stakeholders shaping Volkswagen's business strategy were corporate management and engineering task teams in research and development.

Europe became a global leader in mitigating global warming and climate change. Both the EC and ACEA acted as key external stakeholders providing regulatory and financial incentives for carmakers to invest in diesel technologies. Demand for diesel cars was bolstered by positive media coverage of "clean" diesel vehicles potentially resulting in a 25% reduction in CO₂ emissions from the automotive industry and contributing toward the EU's overall climate goals. As a result, consumers in Europe were given the impression that by buying a diesel car, they were making a personal contribution to addressing climate change.

It has been argued that corporate strategy, economic development, and market competition in developed economies are rooted in technological innovation (Spring, Hughes, Mason, & McCaffrey, 2017; Zhang, 2020). Indeed, companies like Volkswagen regard innovation as a strategic competitive edge (Adner & Kapoor, 2010). The development of the TDI engine made it possible for Volkswagen to take diesel cars from the niche to the mainstream. Governments also recognize the pivotal role that innovation has in economic development (Spring et al., 2017) and use regulations to influence businesses' corporate

behavior and strategy (Penna & Geels, 2015). Lawmakers in Europe co-created value for diesel technological innovation, providing incentives for carmakers to focus on "clean" diesel engines with lower carbon emissions and giving Volkswagen an edge.

Corporate managers in the automotive industry are under intense pressure to choose and develop the most competitive engine design: petrol, diesel, electric, or hybrid. Car manufacturers generally have projects of technology development, but many of these focus on a particular design to dominate a niche market: electric (Tesla), hybrid (Toyota), or diesel (Volkswagen).

According to Don Hillebrand, Director of Energy Systems Research at Argonne National Laboratory, there are three key factors determining engine performance: engine power, fuel efficiency, and pollutant emissions, and "you get to choose two of them" (Ghose, 2015). Engineers can make engines with a better balance of (a) engine power + fuel efficiency; (b) fuel efficiency + pollutant emissions; and (c) emissions + engine power. Technological innovations in internal combustion engines (ICEs) have advanced in both Option 1 and Option 2 but failed in Option 3. Volkswagen's management decided to adopt Option 3, perhaps because US consumers place greater emphasis on engine power versus fuel efficiency.

As the Automobile Association (2020) notes, consumers are faced with a dilemma when making a purchase decision. Their choice of model and fuel type implies a decision about balancing the challenges of climate change and air quality in towns and cities. Volkswagen's

claims about their diesel engines suggested that TDI-equipped models had better engine power and lower emissions, appealing to consumers concerned about climate change.

Volkswagen management placed CSR and sustainability at the core of the strategic agenda for Strategy 2018, as their 2009 and 2010 annual reports make clear. Using CSR-oriented marketing as a pre-emptive strategy to gain a competitive edge has become a general trend of business practice (Brown & Dacin, 1997; Klein & Dawar, 2004; Sen & Bhattacharya, 2001). Volkswagen's strategic efforts broadly fell within the field of sustainable mobility, which Zhang & Yazdani (2014, p. 15) define as "people's ability to access economic and social opportunities while minimising the environmental impact of their movement and activities." The Volkswagen strategy of positioning their diesel vehicles as "clean" cars were key to their efforts to expand dieselization in the US. Their forceful marketing campaign and positive media coverage of diesel cars convinced American consumers of the benefits of purchasing a diesel-equipped vehicle. Car dealerships, media, and consumers all participated in the resultant value co-creation.

As the emission's scandal revealed, however, the engines were not as climate-friendly as the company claimed. Attempts to circumvent emissions regulations are common practice and carmakers have "successfully done so for decades" (Transport & Environment, 2018), but the Dieselgate scandal, which occurred on an unprecedented scale, forced the practice of emission's fraud into the public domain.

The revelation of their efforts to cheat emissions tests contradicted their public sustainability messaging and led to extreme negative brand sentiment, further amplified through global media coverage of the scandal in September 2015. The consumer backlash against Volkswagen was immediate, although it took 3 years for the EPA to obtain incontrovertible proof of the company's unethical behavior from a whistle-blower on the engineering team.

However, the effects of the scandal were not limited to Volkswagen's reputation. The revelations drew attention to the harmful effects of NO_x emissions (Campbell, 2019) and the broader practice of emission's fraud, as well as discrediting the positive press coverage and corporate marketing about diesel cars that had accompanied Volkswagen's push into the US market. A rapid outcome was a loss of consumer confidence in diesel cars in general.

In response to Dieselgate, sales of diesel cars decreased significantly, partly attributed to consumers' increased awareness of the dangers of NO_x and the changes in associated environmental taxation (Campbell, 2019). New regulations were also introduced through the use of diesel cars. For instance, in April 2019, the city of London introduced air quality regulations meaning drivers of diesel vehicles with a Euro5 or older model engine are subject to a £12.50 charge on top of their increased annual road tax (Evans, 2020).

5 | CONCLUSIONS AND IMPLICATIONS

Considering the Dieselgate scandal and its effects on the demand and regulation of diesel vehicles, the studies of CSI need to take into

account not just the behavior of organizations and consumers, but the interactions within the stakeholder ecosystem. Multiple stakeholders are engaged in the co-creation of values that can lead to the growth of supply and demand. Our case analysis shows the critical roles played by regulators, NGOs, and whistle-blowers. Mutual interactions precipitated changes in the market that went beyond Volkswagen to influence broader demand for diesel cars.

Over the long term, the emissions cheating scandal also led to a loss of reputational capital for Volkswagen. The company's management team understands the importance of restoring trust by engaging with constituencies in its stakeholder ecosystem, which they neglected in the past. Current CEO Herbert Diess has proposed recruiting an environmental activist to oversee Volkswagen's environmental strategy (Miller & Campbell, 2020), a highly strategic move which is closely tied to its new corporate strategy Together 2025⁺. This new strategy emphasizes Volkswagen's role in the safety and environmental protection by acting following its stated values: "our highest principles include integrity, the observance of rules and laws, the safety of our customers... [and] open dealing with errors in order to remedy and avoid them in the future" (Volkswagen, 2020).

There are, however, two issues to consider in the strategic implementation of such high principles. First, any organization which has been guilty of CSI needs to understand the difference between brand loyalty and corporate reputation. Both involve efforts to build or restore trust, but they require a different approach (Swalef & Zhang, 2018). Previous research has shown that brand and reputation while overlapping, are two different concepts. As Ettenson and Knowles (2008) argue, a brand is a "customer-centric" concept that companies use to assure customers that their products and services will deliver promised value propositions.

Conversely, reputation is a "company-centric" concept that underpins a business's credibility and integrity within its stakeholder ecosystems. To build a reputation for corporate integrity, companies need to display trustworthy behavior that members of their stakeholder ecosystems will monitor and recognize (Mayer, Davis, & Schoorman, 1995; Swalef & Zhang, 2018). It is only by developing and showing trustworthiness that companies can co-create enduring value with their stakeholders (Gyrd-Jones & Kornum, 2013; Kazadi et al., 2016; Pera et al., 2016).

Volkswagen's Together 2025⁺ strategy is key to its long-term efforts to rebuild its reputation. The strategy aims to "make mobility sustainable for us and future generations" (Volkswagen, 2020). Transport and Environment (2018) contends that the first step of the sustainable mobility transition is a shift to electro-mobility, and in keeping with this, Volkswagen has announced plans to invest 33 billion euros in developing electric vehicles (Miller & Campbell, 2020).

The Dieselgate scandal reveals the tension between the theory and practice of CSR. Companies may formulate effective strategies that rest on a sustainable and responsible business model, but fail in their implementation. Volkswagen created a sustainability board in 2009 as part of its corporate CSR campaign (Volkswagen, 2010) and sold cars equipped with default devices as "clean" vehicles. Its successful development and marketing of TDI diesel models accelerated

the dieselization process in Europe and managed to deploy the “halo effect” on the US market. The Jetta TDI model, benefiting from this “halo effect,” won Green Car of the Year in 2009. Considerations about the strategic benefits of CSR and the halo effect influenced Volkswagen's corporate strategy (Hong & Liskovich, 2015), but rather than motivating healthier behavior, and they led to a concerted strategy to cheat emission's tests.

Our paper has implications for both business theory and managerial practice. Recent research in the emerging field of CSI studies provides fertile ground for further theoretical development (Antonetti, 2020; Grappi et al., 2013; Lange & Washburn, 2012; Pearce & Manz, 2011; Riera & Iborra, 2017). Examining CSI in detail is an opportunity to explore the complex, multi-faceted nature of unethical corporate behavior. Our case study shows that the dieselization process in Europe emerged from value co-creation between Volkswagen, policymakers, and other stakeholders.

In 2018 and 2019, after intense efforts to re-establish trust, Volkswagen grew its market share to become the world's largest car manufacturer (Focus2move, 2020). Its success came in part through its proactive efforts to repair its reputation and re-establish trust within its stakeholder ecosystem. For managerial practice, we see the relevance of companies broadening their perspective when they formulate and implement their strategy, for instance, by including NGOs and critical media. Volkswagen's proposal to develop its future strategy together with an environmental activist is a positive departure from its past behavior and a robust measure to convince stakeholders of its integrity (Miller & Campbell, 2020). However, it is only the start of necessary strategic changes in managerial practice.

A genuine transition to sustainable mobility in the automotive industry, and to sustainability in general, requires managers to make radical changes in their strategy and behavior in consultation with key stakeholders, to ensure that they follow sustainability principles rather than merely paying lip service. Our case study shows that the effects CSI scandals can spill over to affect demand in the broader market, not only for the company responsible for the damage. Creating and sustaining stakeholder ecosystems can enable organizations to co-create value with their stakeholders and at the same time to safeguard the objectives of sustainability, which in turn will prevent organizations from engaging in CSI.

REFERENCES

- Adner, R., & Kapoor, R. (2010). Value creation in innovation ecosystems: How the structure of technological interdependence affects firm performance in new technology generations. *Strategic Management Journal*, 31, 306–333.
- Antonetti, P. (2020). More than just a feeling: A research agenda for the study of consumer emotions following corporate social irresponsibility (CSI). *Australasian Marketing Journal*, 28, 67–70. <https://doi.org/10.1016/j.ausmj.2020.01.005>
- Aouad, A., & Marsat, S. (2018). Do ESG controversies matter for firm value? Evidence from international data. *Journal of Business Ethics*, 151, 1027–1047.
- Automobile Association (2020). Diesel, petrol, electric or hybrid? Retrieved from <https://www.theaa.com/car-buying/fuel-types>.
- BBC News (2020) Volkswagen in ‘dieselgate’ talks with motorists. Retrieved from: <https://www.bbc.co.uk/news/business-50971156>
- Brown, T., & Dacin, P. (1997). The Company and the Product: Corporate Associations and Consumer Product Responses. *Journal of Marketing*, 61, 68–84.
- Bryson, D., & Atwal, G. (2018). Brand hate: The case of starbucks in France. *British Food Journal*, 121(1), 172–182.
- Campbell, P. (2019). “London clean air rules accelerate slump in diesel cars demand”. Retrieved from <https://www.ft.com/content/6378f18a-8dc5-11e9-a1c1-51bf8f989972>
- Carroll, A. (1979). A three-dimensional conceptual model of corporate performance. *Academy of Management Review*, 4, 497–505.
- Chossière, G. P., Malina, R., Ashok, A., Dedoussi, I. C., Eastham, S. D., Speth, R. L., & Barrett, S. R. H. (2017). Public health impacts of excess NOx emissions from Volkswagen diesel passenger vehicles in Germany. *Environmental Research Letters*, 12(3), 034014.
- Coffey, B., & Fryxell, G. (1991). Institutional ownership of stock and dimensions of corporate social performance: An empirical examination. *Journal of Business Ethics*, 10, 437–444.
- Dieselnet (2020). Emission Standards United States: Cars and Light-Duty Trucks. Retrieved from <https://dieselnet.com/standards/us/ld.php>
- Ettenson, R., & Knowles, J. (2008). Don't confuse reputation with brand. *MIT Sloan Management Review*, 49(2), 19–21.
- European Commission (1998). Commission and ACEA agree on CO₂ emission from cars. Brussels: IP/98/734. Retrieved from https://ec.europa.eu/commission/presscorner/detail/en/IP_98_734
- European Commission (2017). Commission presents guidance for Member States to enforce ban on defeat devices. Retrieved from <https://ec.europa.eu/jrc/en/news/commission-presents-guidance-member-states-enforce-ban-defeat-devices>
- Evans, C. (2020). “Should I buy a diesel car? Everything you need to know”. Retrieved from <https://www.whatcar.com/advice/buying/should-i-buy-a-diesel-car-everything-you-need-to-know/n1189>
- Fisher, D. (2015). VW's Diesel Was A Creation Of EU Regulators. *Forbes*, retrieved from: <https://www.forbes.com/sites/danielfisher/2015/10/02/eu-emissions-regulations-made-the-vw-diesel-u-s-regs-brought-it-down/#634aaeda5f2f> (accessed 14 May 2020).
- Focus2move (2020). World Ranking: Global Auto Market 2020. Retrieved from <https://www.focus2move.com/world-car-group-ranking/>
- Ghose, T. (2015). Volkswagen scandal: Why is it so hard to make clean diesel cars? Retrieved from: <https://www.livescience.com/52284-volkswagen-scandal-clean-diesel-challenges.html>.
- Grappi, S., Romani, S., & Bagozzi, R. (2013). Consumer response to corporate irresponsible behavior: Moral emotions and virtues. *Journal of Business Research*, 66, 1814–1821.
- Groom, N., & Krollick, K. (2008). Volkswagen diesel car wins “Green Car of the Year”. Retrieved from <https://www.reuters.com/article/us-autos-greencar-idUSTRE4AJ7S120081120>
- Gyrd-Jones, R., & Kornum, N. (2013). Managing the co-created brand: Value and cultural complementarity in online and offline multi-stakeholder ecosystems. *Journal of Business Research*, 66, 1484–1493.
- Hong, H. & Liskovich, I. (2015). Crime, punishment and the halo effect of corporate social responsibility. NBER working paper series, working paper 21215, Retrieved from <http://www.nber.org/papers/w21215>.
- Kazadi, K., Lievens, A., & Mahr, D. (2016). Stakeholder co-creation during the innovation process: Identifying capabilities for knowledge creation among multiple stakeholders. *Journal of Business Research*, 69, 525–540.
- Klein, J., & Dawar, N. (2004). Corporate social responsibility and consumers' attributions and brand evaluations in a product-harm crisis. *International Journal of Research in Marketing*, 21(3), 203–217.
- Kozinets, R., & Handelman, J. (2004). Adversaries of consumption: Consumer movements, activism, and ideology. *Journal of Consumer Research*, 31(3), 691–704.

- Lange, D., & Washburn, N. (2012). Understanding Attributions of Corporate Social Irresponsibility. *Academy of Management Review*, 37, 300–326.
- Lee, M., Motion, J., & Conroy, D. (2009). Anti-consumption and brand avoidance. *Journal of Business Research*, 62(2), 169–180.
- Lussenhop, J. (2015). *Why do American car buyers shy away from diesel?* Retrieved from: <https://www.bbc.co.uk/news/world-us-canada-34329596>.
- Mayer, R., Davis, J., & Schoorman, D. (1995). An integrative model of organisational trust. *Academy of Management Review*, 20, 709–734.
- McGee, P. (2016). “Volkswagen diesel emissions scandal far from over”. Retrieved from: <https://www.ft.com/content/956928ba-3d49-11e6-9f2c-36b487ebd80a>
- McGee, P. (2017). How VW's cheating on emissions was exposed. Retrieved from <https://www.ft.com/content/103dbe6a-d7a6-11e6-944b-e7eb37a6aa8e>
- McGuire, J., Sundgren, A., & Schneeweis, T. (1988). Corporate social responsibility and firm financial performance. *Academy of Management Journal*, 31, 854–872.
- Miller, J., & Campbell, P. (2020). VW to appoint ‘aggressive’ climate activist to scrutinise policies. Retrieved from <https://www.ft.com/content/213c954a-5748-11ea-a528-dd0f971febbc>.
- Miravete, E., Moral, M., & Thurk, J. (2015). Innovation, emissions policy, and competitive advantage in the diffusion of European diesel automobiles. Retrieved from <https://voxeu.org/article/diffusion-european-diesel-automobiles>
- Murray, K., & Vogel, C. (1997). Using a hierarchy of effects approach to gauge the effectiveness of CSR to generate goodwill towards the firm: Financial versus nonfinancial impacts. *Journal of Business Research*, 38, 141–159.
- Noria Corporation (2008). VW of America moves forward with ‘Strategy 2018’ plan. Retrieved from <https://www.reliableplant.com/Read/15320/vw-of-america-moves-forward-with-'strategy-2018'-plan>
- Pearce, C., & Manz, C. (2011). Leadership centrality and corporate social Ir-responsibility (CSIR): The potential ameliorating effects of self and shared leadership on CSIR. *Journal of Business Ethics*, 102, 563–579.
- Penna, C., & Geels, F. (2015). Climate change and the slow reorientation of the American car industry (1979–2012): An application and extension of the dialectic issue LifeCycle (DILC) model. *Research Policy*, 44, 1029–1048.
- Pera, R., Occhiocupo, N., & Clarke, J. (2016). Motives and resources for value co-creation in a multi-stakeholder ecosystem: A managerial perspective. *Journal of Business Research*, 69, 4033–4041.
- Reypens, C., Lievens, A., & Blazevic, V. (2016). Leveraging value in multi-stakeholder innovation networks: A process framework for value co-creation and capture. *Industrial Marketing Management*, 56, 40–50.
- Riera, M., & Iborra, M. (2017). Corporate social irresponsibility: Review and conceptual boundaries. *European Journal of Management and Business Economics*, 26(2), 146–162.
- Sen, S., & Bhattacharya, C. (2001). Does doing good always lead to doing better? Consumer reactions to corporate social responsibility. *Journal of Marketing Research*, 38, 225–243.
- Spring, M., Hughes, A., Mason, K., & McCaffrey, P. (2017). Creating the competitive edge: A new relationship between operations management and industrial policy. *Journal of Operations Management*, 49, 6–19.
- Swalef, R., & Zhang, M. (2018). Restoring trust through isomorphism at MFIs: Trustworthiness and boundary permeability. In M. Zhang (Ed.), *Trust building and boundary spanning in cross-border management* (pp. 157–180). New York: Routledge: Routledge Studies in Trust Research.
- Transport & Environment. (2018). *CO₂ emissions from cars: The facts*. Brussels: European Federation for Transport and Environment AISBL. <https://www.transportenvironment.org/publications/co2-emissions-cars-facts>.
- Vargo, S., & Lusch, R. (2008). Service-dominant logic: Continuing the evolution. *Journal of the Academy of Marketing Science*, 36, 1–10.
- Vidal, J. (2015). The rise of diesel in Europe: The impact on health and pollution. Retrieved from: <https://www.theguardian.com/environment/2015/sep/22/the-rise-diesel-in-europe-impact-on-health-pollution>.
- Volkswagen (2009). *Driving Ideas*. VW Annual Report 2009. 38436 Wolfsburg, ISSN 058.809.522.20.
- Volkswagen (2010). *Experience D[r]iversity*. Experience D[r]iversity. VW Annual Report 2010. 38436 Wolfsburg, ISSN 158.809.527.20.
- Volkswagen (2020). *Together 2025+*. Retrieved from <https://www.volkswagenag.com/en/group/strategy.html>
- VW US Media (2015). Annual deliveries by Volkswagen Passenger Cars top six million for first time. Retrieved from <https://media.vw.com/en-us/releases/423> (accessed 26 May 2020).
- Wright, R., & Sharman, A. (2015). VW cheated emissions tests, says EPA. Retrieved from <https://www.ft.com/content/5e0b8bcc-5e21-11e5-9846-de406ccb37f2>
- Zhang, M. (2020). Innovation Ecosystems for Sustainability Transition: From Policy Intervention to Stakeholder Coalition. Retrieved from <https://ieeexplore.ieee.org/search/searchresult.jsp?newsearch=true&queryText=tems%20isie%202019&highlight=true&returnFacets=ALL&returnType=SEARCH&matchPubs=true&pageNumber=2>
- Zhang, M., & Yazdani, B. (2014). Paradigm change of mobility in the twenty-first century. *Journal of Sustainable Mobility*, 1(1), 9–18.

AUTHOR BIOGRAPHIES

Michael Zhang is Reader (Associate Professor) in International Strategy at Nottingham Business School, Nottingham Trent University, UK.

Glyn Atwal is Associate Professor at Burgundy School of Business, France.

Maya Kaiser is an Adjunct Professor at a number of leading business schools in France.

How to cite this article: Zhang M, Atwal G, Kaiser M. Corporate social irresponsibility and stakeholder ecosystems: The case of Volkswagen Dieselgate scandal. *Strategic Change*. 2021;30:79–85. <https://doi.org/10.1002/jsc.2391>